Assignment 10 Solution

Q.1. What effect does the contrast intensifier have on membership values based on the given options?

(a) Increases values above 0.5 and diminishes values below 0.5.
(b) Diminishes values above 0.5 and increases values below 0.5.
(c) Increases values both above and below 0.5.
(d) Diminishes values both above and below 0.5.
(e) None of the above.

Solution: (a)

Q.2. Let fuzzy set A is defined with the universe of discourse $X = \{1,2,3,4\}$ given as below.

$$A = 0.1/1 + 0.3/2 + 1.0/3 + 0.5/4$$

What will be the membership function value at generic variable value 2 for the fuzzy set obtained by $INT(A)$?

(a) 1.00
(b) 0.02
(c) 0.18
(d) 0.50

Solution: (c)

$$A = 0.1/1 + 0.3/2 + 1.0/3 + 0.5/4$$

For $x = 1 \Rightarrow \mu_A(x) = 0.1 \Rightarrow 0.0 \leq \mu_A(x) \leq 0.5$:

$$2A^{(2)} = 2(0.1)^{(2)}/1 = 2 \times 0.01/1 = 0.02/1$$

For $x = 2 \Rightarrow \mu_A(x) = 0.3 \Rightarrow 0.0 \leq \mu_A(x) \leq 0.5$:

$$2A^{(2)} = 2(0.3)^{(2)}/2 = 2 \times 0.09/2 = 0.18/2$$

For $x = 3 \Rightarrow \mu_A(x) = 1.0 \Rightarrow 0.5 \leq \mu_A(x) \leq 1.0$:

$$\neg 2(\neg A)^{(2)} = \neg 2(1 - 1.0)^{(2)}/3 = (1 - 0.00)/3 = 1.00/3$$

For $x = 4 \Rightarrow \mu_A(x) = 0.5 \Rightarrow 0.0 \leq \mu_A(x) \leq 0.5$:

$$2A^{(2)} = 2(0.5)^{(2)}/2 = 2 \times 0.25/2 = 0.50/2$$

$$INT(A) = 0.02/1 + 0.18/2 + 1.00/3 + 0.50/4$$

Q.3. For the fuzzy set defined in Q.2., What will be the membership function value at generic variable value 4 for the fuzzy set obtained by $INT(A)$?

(a) 1.00
(b) 0.02
(c) 0.18
(d) 0.50

Solution: (d)

Q.4. For the fuzzy set defined in Q.2., What will be the membership function value at generic variable value 1 for the fuzzy set obtained by $INT(A)$?

(a) 1.00
(b) 0.02
(c) 0.18
(d) 0.50

Solution: (b)

Q.5. For the fuzzy set defined in Q.2., What will be the membership function value at generic variable value 3 for the fuzzy set obtained by $INT(A)$?

(a) 1.00
(b) 0.02
(c) 0.18
(d) 0.50

Solution: (a)

Q.6. For the fuzzy set defined in Q.2., What will be the completement of $INT(A)$?

(a) 0.02/1 + 0.18/2 + 1.00/3 + 1.00/4
(b) 0.98/1 + 0.18/2 + 1.00/3 + 0.50/4
(c) 0.98/1 + 0.82/2 + 1.00/3 + 0.50/4
(d) 0.98/1 + 0.82/2 + 0.00/3 + 0.50/4

Solution: (d)

Q.7. For a fuzzy IF-THEN rule defined using "IF x is A, THEN y is B", which of the following statements are true?

(a) "x is A" is called as consequence and "y is B" is called as antecedent.
(b) "x is A" is called as antecedent and "y is B" is called as consequent.
(c) "x is A" is called as conclusion and "y is B" is called as premise.
(d) "x is A" is called as conclusion and "y is B" is called as consequent.

Solution: (b)

Q.8. What is the primary function of a fuzzifier in a fuzzy logic system?

(a) Convert fuzzy values to crisp inputs
(b) Convert crisp inputs to fuzzy values
(c) Evaluate membership functions
(d) Implement logical operations in fuzzy logic

Solution: (b)

Q.9. What is the primary purpose of a defuzzifier in a fuzzy logic system?

(a) Convert fuzzy values to crisp outputs
(b) Convert crisp inputs to fuzzy values
(c) Determine membership functions
(d) Perform fuzzy inference

Solution: (a)

Q.10. What is the main principle behind fuzzy reasoning in fuzzy logic systems?

(a) Utilizing precise mathematical computations for decision-making
(b) Handling imprecise or uncertain data using linguistic variables
(c) Applying Boolean logic operations for inference
(d) Implementing crisp input-output mappings

Solution: (b)